# Sparsity-Based Clustering for Large Hyperspectral Remote Sensing Images

Han Zhai, *Member, IEEE*, Hongyan Zhang, *Senior Member, IEEE*, Liangpei Zhang, *Fellow, IEEE,* and Pingxiang Li, *Member, IEEE*

***Abstract*— Hyperspectral image (HSI) clustering is extremely challenging because of the complexity of the image structure. Recently, the subspace clustering algorithms have achieved competitive performance for HSIs. However, these methods generally are computationally complex and time-and-memory-consuming, given their reliance on large-scale adjacency matrix learning and graph segmentation, which limits their application to large HSIs and reduces their attractiveness in real applications. In this article, in view of this, two novel sparsity-based clustering algorithms are proposed for large HSIs, named sparse coding-based clustering (SCC) and joint SCC (JSCC). To the best of our knowledge, we are the first to use the sparse representation recovery residual to cluster HSIs. Based on a structured dictionary constructed by *k*-means and *k*-nearest neighbor (KNN), an SCC model is constructed to cluster HSIs according to the recovery residual minimization criterion. By dealing with a pixel-wise sparse recovery problem instead of the large-scale graph optimization problem of the whole image, the computational complexity and the time-and-memory cost are reduced to a large degree, which makes sense for practical applications. Then, by introducing the super-pixel neighborhood, a JSCC model is constructed to better explore the interpixel correlation of HSIs and further improve the clustering performance. The proposed algorithms were verified on three widely used HSIs. All the three experiments confirm the effectiveness of the proposed algorithms, which can be considered as competitive tools for use with large HSIs.**

***Index Terms*— Clustering, hyperspectral image (HSI), joint sparse coding, recovery residual, sparse coding.**

## I. Introduction

HYPERSPECTRAL sensors can image a spatial area of interest at the nanoscale spectral resolution, with each pixel corresponding to a nearly continuous spectrum, which can capture the diagnostic features and subtle differences of the various materials to facilitate fine classification and information extraction [1]–[5]. Clustering is a common unsupervised hyperspectral image (HSI) interpretation technique, which does not require any labeled samples. It has a wide range of applications in many fields, such as forestry studies [6], precision agriculture [7], and environmental monitoring [8]. The aim of HSI clustering is to group similar pixels from the same class while separating the dissimilar pixels from different classes. However, because of the complex internal structure of HSIs and the large differences between the spectra of pixels from the same class, HSI clustering has always been a very challenging task.

HSI clustering has been around for decades, and many methods have been proposed [9], [10]. According to the theoretical basis and the working mechanism, HSI clustering mainly includes the following five categories: 1) centroid-based models; 2) density-based models; 3) probability-based models; 4) bionics-based models; and 5) graph-based models:

The centroid-based models assume that pixels in the feature space can be grouped into several clusters with the *n*-sphere structure, by constantly updating the centroid locations and the pixel partition modes, according to the minimization criterion of the total squared distance error. Typical examples are *k*-means [11], fuzzy *c*-means (FCM) [12], FCM incorporating spatial-contextual information (FCM-S1) [13], FCM with spatial membership relations (FCMSMRs) [14], FCM with adaptive local information (ADFLICM) [15], and FCM incorporating local double neighborhood information (FLDNICM) [16]. The density-based models assume that cluster generally follows a dense distribution in the relative sparse high-dimensional feature space of HSIs and group pixels by calculating their density and relative distances, such as the clustering by fast search and find of density peaks (CFSFDPs) [17], density analysis ensemble (DAE) [18], unsupervised HSI clustering based on adaptive density (UCAD) [19], nonlinear diffusion-based active learning and clustering (NDALC) [20], and graph watershed clustering based on nearest neighbors (GWCNNs) [21]. The probability-based models assume that hyperspectral pixels satisfy certain mixed probability models and group them according to the probability maximization criterion. The representative methods include the Gaussian mixture model (GMM) [22], independent component analysis model (ICA) [23], GMM with Markov random fields (GMM-MRFs) [24], and latent Dirichlet-distribution-based clustering (LDDC) [25]. The bionics-based models simulate the internal structure

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of HSIs with certain biological models and partition pixels through evolutionary computation. Examples of such methods are the artificial immune clustering (AIC) [26], artificial-immune-network-based clustering (AINC) [27], artificial clustering by DNA computing and spectral matching (ACDCSM) [28], and extreme learning-machine-based clustering (ELMC) [29]. The graph-based models simulate the relationships between pixels with a similarity graph and segment the HSI by certain graph-cut techniques, such as the spectral clustering (SC) [30], sequential SC (S-SC) [31], Schrodinger eigenmap-based clustering (SEC) [32], and graph-based nonlocal total variation clustering (NLTVC) [33].

However, the pixel distribution in the high-dimensional feature space is relatively sparse and uniform for HSIs, without any obvious pattern to follow, caused by the complex internal structure and the large spectral variability, which makes the assumptions of most of the above methods not be well-satisfied [10], [34]. Therein, the clustering performance of these methods is generally limited to a large degree, which leads to an unsatisfactory clustering result.

The recently developed subspace clustering algorithms are capable of alleviating these problems to some degree, by more accurately modeling the HSIs with subspaces [9], [10], [34], and they can generally obtain good results. Specifically, such methods model pixels belonging to the same category with a specific subspace, exploring the underlying adjacency between hyperspectral pixels by self-representation learning, with the HSI itself used as the dictionary. A typical example is the sparse subspace clustering (SSC) [9], [35], which has shown a large potential for HSIs. Considering the absence of spatial information in the SSC model, several spectral–spatial enhanced versions have been constructed to further improve the clustering accuracy, by integrating the spatial-contextual information into the SSC model, such as the SSC incorporating spatial information (SSC-S) [9], $\ell_2$-norm regularized SSC (L2-SSC) [10], SSC with 3-D edge-preserving-filtering (SSC-3DEPF) [36], SSC based on new adjacency matrix [37], and super-pixel-based SSC with Markov random field (SSC-MRF) [38]. Besides, to better explore the global structure information, a set of low-rank clustering methods have been developed, such as the low-rank representation (LRR) [39], low-rank subspace clustering (LRSC) [40], and kernel sparse and LRSC (KSLRSC) [41].

Unfortunately, the subspace clustering methods commonly suffer from a high computational complexity and are time-and-memory consuming [9], [10], which makes them less suitable for large HSIs and less attractive in real applications. Thus, some online techniques have been developed to overcome the obstacles caused by the large-scale adjacency matrix learning, such as the online dictionary learning for sparse coding [42], online LRSC by basis dictionary pursuit [43], and online hyperplanes clustering [44]. In addition, some fast graph-cut techniques are used to reduce the complexity caused by the large-scale graph segmentation, such as the fast LRSC [45], fast clustering based on semidefinite programs of graph cut [46], and landmark-based SC [47]. Such methods reduce the computational complexity to a large degree by means of online learning or fast graph cut and have been widely used in the computer vision field, but with few applications in the HSI clustering field.

Recently, many clustering methods have been proposed for large HSIs. The most representative approach is the anchor-graph-based clustering, such as the fast SC with anchor graph (FSCAG) [48] and the scalable-graph-based clustering with non-negative relaxation (SGCNR) [49]. Through selecting a few anchor pixels to construct a much smaller sized similarity graph, the computational complexity is greatly reduced, which makes these methods very efficient and more suitable for large HSIs. However, the anchors are generally randomly selected, ignoring the interaction between pixels, which reduces the accuracy in mining the underlying similarity between pixels to a large degree, thereby limiting their clustering performance.

As we know, sparse coding can more accurately exploit the interaction between pixels by promoting each target pixel to be reconstructed by only a few highly correlated atoms, which has been successfully applied to HSI classification and has exhibited a powerful information extraction capability to yield accurate classification results [50]–[52]. In addition, by taking advantage of the high spatial correlation of HSIs, many enhanced joint sparse-coding-based classifiers have been developed to further improve the classification accuracy, which obtain very competitive classification performance [53]–[55].

In this article, in view of this, by introducing the sparse representation theory to HSI clustering, two novel sparsity-based clustering methods are proposed for use with large HSIs. The contributions of this article are threefold. First, to the best of our knowledge, we are the first to address the HSI clustering problem using sparse coding and recovery residual. Considering that there are no labeled samples or prior knowledge available, $k$-means is used to precluster the HSIs to obtain the cluster centroids, and then, based on the $k$-nearest neighbor (KNN) strategy, a meaningful structured dictionary is constructed for sparse coding. Second, faced with the computational obstacles of subspace clustering, using the structured dictionary, a sparse-coding-based clustering (SCC) model is constructed to cluster HSIs according to the recovery residual minimization criterion. Through replacing the large-scale graph optimization problem of the whole image by a pixel-wise sparse recovery problem, the computational complexity and the time-and-memory cost can be reduced to a large degree, which makes sense for practical applications and large HSI clustering. Third, considering the lack of spatial information in the SCC model and the weak homogeneity of regular neighborhood in the traditional joint sparsity models, based on a super-pixel neighborhood, a joint SCC (JSCC) model is constructed to further improve the clustering performance and enhance the smoothness of the clustering results, by taking advantage of the spectral and spatial domain information of HSIs simultaneously and more finely exploring the interpixel correlation.

The structure of this article is as follows. Section II simply introduces the related work. The proposed sparsity-based clustering algorithms are described in detail in Section III. Section IV presents and analyzes the experimental results based on three well-known HSI images. Section V summarizes

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this article and points out some possible future research directions.

## II. Related Work

In the recent years, many sparse-representation-based classification (SRC) methods have been developed for HSIs and have achieved competitive classification performance [50]–[52]. Such methods are based on the assumption that each pixel can be sparsely recovered by only a few highly correlated atoms in the over-completed dictionary.

Generally, the 3-D HSI cube can be expressed as a 2-D matrix $\mathbf{Y} = [\mathbf{y}_1, \mathbf{y}_2, \ldots, \mathbf{y}_{MN}] \in \mathbb{R}^{D\times MN}$, where $\mathbf{y}_i \in \mathbb{R}^D$ denotes the $i$th pixel, and $D$ and $MN$ represent the number of bands and the number of pixels, respectively. First, with the help of the training samples, an over-completed dictionary can be directly constructed, by constructing a subdictionary for each class with the training samples and sorting the subdictionaries by label values to generate a structured dictionary. Specifically, the structured dictionary is denoted by $\mathbf{D} = [\mathbf{D}_1, \mathbf{D}_2, \ldots, \mathbf{D}_c] \in \mathbb{R}^{D\times N_c}$, where $\mathbf{D}_i$ denotes the $i$th subdictionary for the $i$th class, $c$ means the number of classes, and $N_c$ is the number of atoms.

Then, given the structured dictionary $\mathbf{D}$, the SRC methods deal with the classification task by considering it as a sparse recovery problem, as given in the following equation, to obtain an associated sparse coefficient $\mathbf{c}$ for each target pixel $\mathbf{y}$ [50]–[52]:

$$\arg\min \|\mathbf{c}\|_0 \ s.t. \mathbf{Dc} = \mathbf{y} \tag{1}$$

where $\|\cdot\|_0$ is the $\ell_0$-norm, which counts the number of nonzero items of the coefficient, and $s$ is the predefined sparsity level. As is well-known, the $\ell_0$-norm usually leads to an "NP-hard" problem. In practice, it can be solved by two different kinds of algorithms, that is, greedy pursuit algorithms [56], [57] or relaxed basis pursuit algorithms [58], [59].

Finally, based on the obtained sparse coefficient $\mathbf{c}$, the SRC methods extract discriminative information from the recovery residual and determine the label of $\mathbf{y}$ by minimizing the recovery residual [50]–[52]:

$$\begin{aligned}\text{label}(\mathbf{y}) &= \arg\min_{j=1,\ldots,c} r^j(\mathbf{y}) \\ &= \arg\min_{j=1,\ldots,c} \|\mathbf{D}^j\mathbf{c}^j - \mathbf{y}\|_2, \quad j = 1, \ldots, c\end{aligned} \tag{2}$$

where $r^j(\mathbf{y})$ is the $j$th residual between the sparse recovery from the structured dictionary and the target pixel [50], $\mathbf{D}^j$ is the $j$th subdictionary, and $\mathbf{c}^j$ is the $j$th portion of the sparse coefficient $\mathbf{c}$.

Then, considering the lack of spatial information in the SRC methods, a set of joint SRC (JSRC) methods are developed to further improve the classification performance, by opening a local window to integrate the spatial neighborhood information to help classify the central pixel [50], [53]–[55].

With the structured dictionary $\mathbf{D}$, the JSRC methods can be formulated as follows:

$$\arg\min \|\mathbf{DC} - \mathbf{X}\|_F \quad \text{s.t. } \|\mathbf{C}\|_{\text{row},0} \le s \tag{3}$$

where $\|\cdot\|_F$ is the Frobenius norm, $\|\cdot\|_{\text{row},0}$ is the row-sparse norm, which counts the number of nonzero rows of the matrix, $\mathbf{X} \in \mathbb{R}^{D\times T}$ denotes the pixels in the local window, and $\mathbf{C} \in \mathbb{R}^{N_c\times T}$ denotes the associated sparse coefficients, where $T$ is the number of neighboring pixels. Similarly, (3) is also "NP-hard," which can be solved by the simultaneous orthogonal matching pursuit (SOMP) algorithm [60], [61] or the simultaneous subspace pursuit algorithm [62], [63].

Then, based on the obtained coefficient matrix $\mathbf{C}$, similar to SRC, the label of the central pixel $\mathbf{X}_c$ can be determined by minimizing the joint recovery residual, as given in the following equation:

$$\text{label}(\mathbf{X}_c) = \arg\min_{j=1,\ldots,c} r^c(\mathbf{X}) = \arg\min_{j=1,\ldots,c} \|\mathbf{D}^j\mathbf{C}^j - \mathbf{X}\|_F \tag{4}$$

where $\mathbf{C}^j$ is the $j$th portion of the coefficient matrix $\mathbf{C}$.

## III. Sparsity-Based Clustering for Large Hyperspectral Images

In this section, two sparsity-based clustering models are constructed for use with large HSIs. First, based on the structured dictionary, a novel SCC model is constructed to explore the underlying adjacency between hyperspectral pixels by sparse coding and partition pixels by minimizing the recovery residual. It is then extended to the JSCC model to incorporate the wealthy spatial information of HSIs, to further improve the clustering performance.

### A. Structured Dictionary Construction

As is well-known, dictionary is the basis of sparse coding, and its quality directly affects the effectiveness of sparse recovery. Different from classification, there are neither labeled samples nor prior knowledge available for clustering. Therefore, how to construct a high-quality dictionary for sparse coding to comply with the working mechanism of clustering and how to make the associated sparse coefficient meaningful to contain the underlying ownership information of each target pixel are important but challenging tasks.

At this point, a natural idea is to fully explore the correlation between hyperspectral pixels to select representative atoms for each category to construct a structured dictionary, with all classes included and the atoms of the same class grouped together. In this way, each pixel can be effectively reconstructed based on the dictionary, and its category information can be directly learned by analyzing the structural characteristics of the associated coefficient. Considering that cluster centroids generally have good separability and are from different classes [11], [12], it is reasonable to precluster the HSIs to extract the cluster centroids as the indicators for atom selection. In this article, the classical $k$-means model is used to efficiently partition the HSIs to obtain the cluster centroids. Then, the KNN strategy with Euclidean distance measurement is used, which can effectively measure the similarity between hyperspectral pixels and is widely used in HSI interpretation [64], [65], to select the most similar $k$ atoms to the centroids, to construct the class subdictionaries. Finally, by combining the class subdictionaries in sequence,

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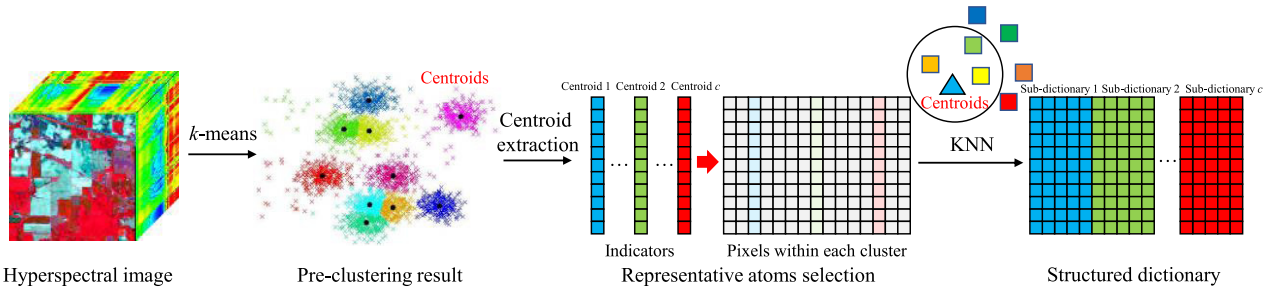



Fig. 1. Scheme of the structured dictionary construction.

a structured dictionary can be generated. The process of structured dictionary construction is depicted in Fig. 1.

To be specific, the structured dictionary can be expressed as

$$\mathbf{A} = \left[\underbrace{\mathbf{a}_1, \ldots, \mathbf{a}_d}_{\mathbf{A}^1}, \underbrace{\mathbf{a}_{d+1}, \ldots, \mathbf{a}_{2d}}_{\mathbf{A}^2}, \cdots \underbrace{\mathbf{a}_j, \ldots, \mathbf{a}_K}_{\mathbf{A}^c}\right] \in \mathbb{R}^{D\times K}$$

with $d$ atoms selected for each class, where $\mathbf{a}_i$ is the selected representative pixel, $\{\mathbf{A}^j\}_{j=1,\ldots,c}$ is the $j$th subdictionary, and $K$ is the number of atoms, with $K = d \times c$.

### B. Sparse-Coding-Based Clustering for HSIs

Faced with the computational obstacles of subspace clustering, a new way of data mining for HSI clustering is explored in this article. Considering the powerful ability of sparse coding in information extraction, and inspired by the great success of the sparsity-based classifiers [50]–[52], we introduce the sparse representation theory to HSI clustering and construct an SCC model for use with large HSIs. Based on the compressed sensing (CS) theory, i.e., most natural signals can be sparsely reconstructed with the help of an over-complete dictionary, SCC explores the underlying adjacency between pixels by sparsely recovering each pixel. That is, through representing each pixel with only a few highly correlated atoms in the structured dictionary, the underlying ownership of the target pixel can be recovered from the patterns and structural characteristics of the associated sparse coefficients. Different from the subspace clustering methods, the SCC deals with a pixel-wise sparse recovery problem instead of the large-scale adjacency matrix learning and graph segmentation problems of the whole image. Accordingly, the computational complexity and the time-and-memory cost are reduced to a large degree, which is favorable for practical applications and large HSI clustering.

The SCC inherits the subspace idea from the subspace clustering models to relieve the large spectral variability (caused by the changes in imaging conditions). Specifically, for each hyperspectral pixel $\mathbf{y}_i \in \mathbb{R}^D, i = 1, \ldots, MN$, it can be seen as lying in the union of $c$ subspaces. Based on the structured dictionary $\mathbf{A}$, pixel $\mathbf{y}_i$ can be approximated as follows:

$$\begin{aligned}\mathbf{y}_i &= \mathbf{A}^1\mathbf{c}^1 + \mathbf{A}^2\mathbf{c}^2 + \cdots + \mathbf{A}^c\mathbf{c}^c + \mathbf{n}_i \\ &= [\mathbf{A}^1, \cdots \mathbf{A}^c]\begin{bmatrix}\mathbf{c}^1 \\ \vdots \\ \mathbf{c}^c\end{bmatrix} + \mathbf{n}_i = \mathbf{A}\mathbf{c}_i + \mathbf{n}_i \end{aligned} \quad (5)$$

where $\mathbf{c}^j \in \mathbb{R}^{d\times 1}, j = 1, \cdots, c$ is the sparse vector corresponding to the $j$th subdictionary $\mathbf{A}^j \in \mathbb{R}^{D\times d}$; $\mathbf{c}_i \in \mathbb{R}^{K\times 1}$ is the corresponding sparse coefficient of pixel $\mathbf{y}_i$, which is formed by concatenating the $c$ sparse vectors $\{\mathbf{c}^j\}_{j=1,\cdots,c}$; and $\mathbf{n}_i \in \mathbb{R}^D$ is the coding error. SCC then invokes the sparse constraint to obtain a robust solution and promotes each target pixel to be represented by only a few atoms from its own subspace.

Specifically, the constructed SCC model can be formulated as:

$$\arg\min_{\mathbf{c}_i} \|\mathbf{c}_i\|_0 \quad \text{s.t. } \mathbf{y}_i = \mathbf{A}\mathbf{c}_i + \mathbf{n}_i, \quad i = 1, \cdots, MN \quad (6)$$

where $\mathbf{n}_i$ denotes the coding error. By constraining the sparsity level of the coefficient, the sparse optimization problem (6) can be relaxed to the following equation:

$$\arg\min_{\mathbf{c}_i} \|\mathbf{A}\mathbf{c}_i - \mathbf{y}_i\|_2 \quad \text{s.t. } \|\mathbf{c}_i\|_0 \le s, \quad i = 1, \cdots, MN \quad (7)$$

where $\|\cdot\|_2$ is the $\ell_2$-norm, and $s$ is the predefined sparsity level. In this article, the orthogonal matching pursuit (OMP) algorithm is used to solve the above "NP-hard" optimization problem [56], [57].

After obtaining the sparse coefficient $\mathbf{c}_i$, the thematic information of the target pixel $\mathbf{y}_i$ can be achieved by the characteristics of the sparse coefficient, as in (8). Each pixel is partitioned into the cluster with the minimum residual:

$$\text{label}(\mathbf{y}_i) = \arg\min_{j=1,\ldots,c} \left\|\mathbf{A}^j\mathbf{c}_i^j - \mathbf{y}_i\right\|_2, \quad j = 1, \ldots, c \quad (8)$$

where $\mathbf{A}^j$ is the $j$th subdictionary and $\mathbf{c}_i^j$ is the $j$th portion of the sparse coefficient $\mathbf{c}_i$.

The pseudocode of the SCC algorithm is provided in Algorithm 1, with the working scheme shown in Fig. 2.

### C. Joint-Sparse-Coding-Based Clustering for HSIs

It should be noted that SCC ignores the abundant spatial information of HSIs, which limits the discriminability of the model to some degree and makes it easily affected by salt-and-pepper noise. As we know, the neighboring pixels generally have similar material compositions, and thus sharing similar spectral characteristics [10], [66]. Therefore, it is necessary to take into account the interpixel correlation of HSIs in the sparsity model, to promote the spatial smoothness across neighboring pixels and enhance the spatial homogeneity of the final cluster map. Based on this fact, motivated by the superiority of joint sparse coding [53]–[55], we extend the

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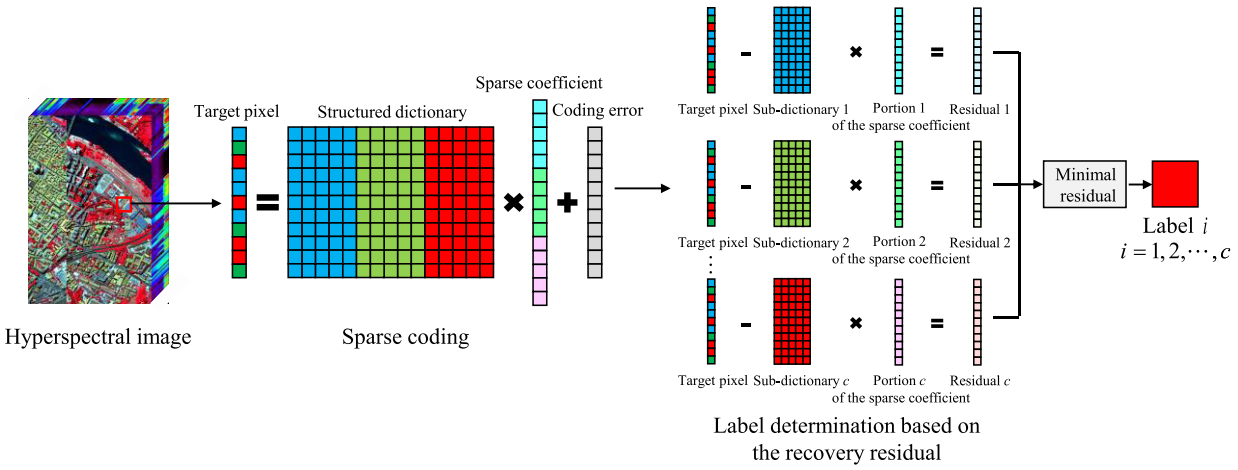



Fig. 2. Working scheme of SCC.

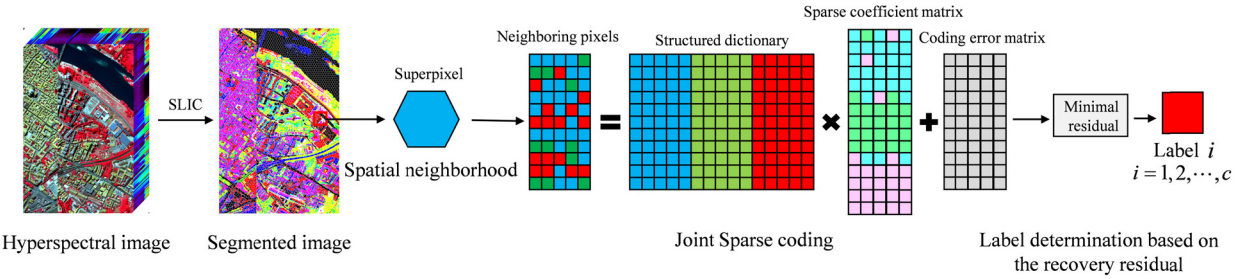



Fig. 3. Working scheme of JSCC.

**Algorithm 1** Sparse-Coding-Based Clustering (SCC) for Hyperspectral Imagery

**Input:**

1) HSI data matrix $\mathbf{Y} = [\mathbf{Y}_1, \mathbf{Y}_2, \ldots, \mathbf{Y}_{MN}]$, $\mathbf{Y} \in \mathbb{R}^{D \times MN}$;
2) Parameters: the number of clusters $c$, the number of atoms for each category $d$, and the sparsity level of the coefficient $s$.

**Main algorithm:**

(1) Construct the structured dictionary $\mathbf{A}$;
(2) Normalize each column of the HSI data matrix $\mathbf{Y}$;
(3) Solve the problem (7) with the OMP algorithm to obtain the sparse coefficient $\mathbf{c}_i$;
(4) Get the labels of pixels based on the minimum residual;

**Output:**

The clustering results with labels.

SCC model to the JSCC model, to integrate the rich spatial information contained in HSIs into the sparsity model to further improve the clustering performance.

JSCC explores the latent interpixel correlation by forcing pixels in a small neighborhood to share the same sparse basis and their associated coefficients to share a common sparsity pattern. It means that the neighboring pixels can be sparsely approximated by the common atoms in the structured dictionary. Specifically, pixels in a small neighborhood $N_p$, $p = 1, \ldots, A$, i.e., $\mathbf{Y}_p = [\mathbf{y}_1, \mathbf{y}_2, \ldots, \mathbf{y}_{T_p}] \in \mathbb{R}^{D \times T_p}$, can be jointly coded as given in the following equation, where $\mathbf{Y}_p$ denotes the $p$th neighborhood of the HSI, $T_p$ represents the number of pixels within the neighborhood, and $A$ is the number of neighborhoods:

$$\begin{aligned}\mathbf{Y}_p &= [\mathbf{y}_1, \mathbf{y}_2, \ldots, \mathbf{y}_{T_p}] \\ &= [\mathbf{A}\mathbf{c}_1 + \mathbf{n}_1, \mathbf{A}\mathbf{c}_2 + \mathbf{n}_2, \ldots, \mathbf{A}\mathbf{c}_{T_p} + \mathbf{n}_{T_p}] \\ &= \mathbf{A}\underbrace{[\mathbf{c}_1, \mathbf{c}_2, \ldots, \mathbf{c}_{T_p}]}_{\mathbf{C}_p} + \mathbf{N}_p = \mathbf{A}\mathbf{C}_p + \mathbf{N}_p \end{aligned} \tag{9}$$

where $\mathbf{C}_p \in \mathbb{R}^{K \times T_p}$ and $\mathbf{N}_p \in \mathbb{R}^{D \times T_p}$ denote the corresponding sparse coefficient matrix and the coding error matrix, respectively. It should be noted that as the sparse coefficients share similar sparse patterns, $\mathbf{C}_p$ should be row-sparse.

Considering that the purity of neighboring pixels cannot be well-guaranteed by the regular neighborhood in the traditional joint sparsity models, especially for the heterogeneous regions, in this article, a super-pixel neighborhood is used for joint sparse coding to more finely explore the spatial-contextual information. Specifically, the widely used simple linear iterative clustering (SLIC) method is used to adaptively acquire the homogeneous super-pixel neighborhoods, based on the first three principal components (PCs) of the HSI [67], [68]. With the help of super-pixels, the purity and homogeneity of the neighboring pixels can be better guaranteed and the joint coding precision can be effectively improved, which is

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**Algorithm 2** Joint-Sparse-Coding-Based Clustering (JSCC) for Hyperspectral Imagery

**Input:**

1) HSI data matrix $\mathbf{Y} = [\mathbf{Y}_1, \mathbf{Y}_2, \ldots, \mathbf{Y}_{MN}], \mathbf{Y} \in \mathbb{R}^{D\times MN}$;
2) Parameters: the number of cluster $c$, the number of atoms for each category $d$, the sparsity level of the coefficient $s$, the number of super-pixels $A$.

**Main algorithm:**

(1) Construct the structured dictionary $\mathbf{A}$;
(2) Normalize each column of the HSI data matrix $\mathbf{Y}$;
(3) Solve the problem (11) with the SOMP algorithm to obtain the associated coefficient matrix $\mathbf{C}_p$;
(4) Get the labels of pixels based on the minimum residual;

**Output:**

The clustering results with labels.

favorable for improving the clustering accuracy. In practice, the SLIC segmentation can affect the final clustering accuracy of the JSCC model by adjusting the number of super-pixels, i.e., parameter $A$. A more detailed introduction of the SLIC method is available in [67].

Specifically, the constructed JSCC model can be formulated as:

$$\arg\min_{\mathbf{C}_p} \|\mathbf{C}_p\|_{\text{row},0} \quad \text{s.t. } \mathbf{Y}_p = \mathbf{A}\mathbf{C}_p + \mathbf{N}_p, \quad p = 1,\ldots, A. \tag{10}$$

In fact, the joint sparse recovery problem (10) is equivalent to the following equation:

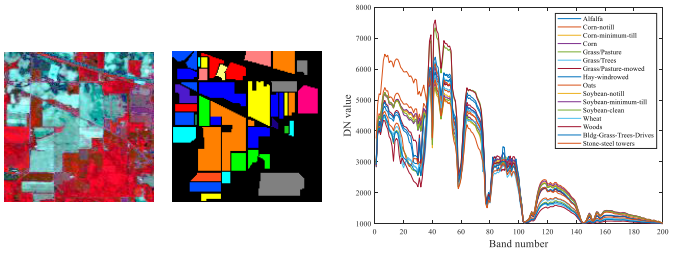
$$\arg\min_{\mathbf{C}_p} \|\mathbf{A}\mathbf{C}_p - \mathbf{Y}_p\|_F \quad \text{s.t. } \|\mathbf{C}_p\|_{\text{row},0} \le s, \quad p = 1, \ldots, A. \tag{11}$$

In this article, the widely used SOMP algorithm is used to solve problem (11) [60], [61]. Then, based on the learned sparse coefficient matrix $\mathbf{C}_p$, the class information of the pixels can then be obtained based on the recovery residual. Similarly, the label of the pixel can be determined directly by the minimum residual criterion, as introduced in Section III-B. The JSCC model is summarized in Algorithm 2, and its working scheme is illustrated in Fig. 3.

## IV. Experiment and Analysis

### A. Experimental Settings

In this section, the performance of the proposed SCC and JSCC algorithms was systematically evaluated, with several representative and state-of-the-art comparative methods used: the FCM method [12], the FCM-S1 method [13], the SSC-S method [9], the L2-SSC method [10], the LRSC method [40], the FSCAG method [48], and the SGCNR method [49]. Specifically, FSCAG and SGCNR are two recently developed advanced methods for use with large HSIs. In the experiments, the cluster number and the thematic information of the cluster map were artificially determined through visual interpretation, which is widely used in HSI clustering [9], [10], [26]. For each




(a) (b) (c)

Fig. 4. Indian Pines image: (a) false-color image (R: 40, G: 30, and B: 20), (b) ground truth, and (c) mean spectra of the 16 classes.

TABLE I
Parameter Settings of Different Clustering Methods for the Indian Pines Image

| Method | Parameters |
|---|---|
| FCM | $c = 16$, $e = 2$ |
| FCM-S1 | $c = 16$, $e = 2$, $\alpha = 0.2$ |
| SSC-S | $c = 16$, $\lambda = 9.82\times10^{-7}$, $\alpha = 9\times10^{3}$, $S = 3$ |
| L2-SSC | $c = 16$, $\lambda = 1.10\times10^{-6}$, $\alpha = 4\times10^{3}$ |
| LRSC | $c = 16$, $\lambda = 10$ |
| SGCNR | $c = 16$, $m = 500$, $k = 5$, $\rho$=1.1, $\mu = 0.1$ |
| FSCAG | $c = 16$, $m = 500$, $k = 5$, $\alpha = 0.6$ |
| SCC | $c = 16$, $d = 20$, $s = 5$ |
| JSCC | $c = 16$, $A = 212$, $d = 20$, $s = 4$ |

$c$: the number of clusters; $e$: fuzzy exponential; $\alpha$: spatial regularization parameter; $\lambda$: regularization parameter; $m$: the number of anchors; $k$: the number of neighbors; $\rho$: penalty parameter; $\mu$: penalty parameter; $d$: the number of atoms for each category; $s$: the sparsity level of the coefficient; $A$: the number of super-pixels; $S$: the size of the local window.

method, the parameters had been fine-tuned to be optimal. For FSCAG and SGCNR, we referred to the original articles for the parameter settings [48], [49].

To thoroughly evaluate the effectiveness of the proposed algorithms, three well-known HSIs were selected, with the image size varying from 21 025 to 783 640 pixels: 1) the Indian Pines image, acquired by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor; 2) the Salinas image, also acquired by the AVIRIS sensor; and 3) the Pavia Center image, collected by the Reflective Optics System Imaging Spectrometer System (ROSIS) sensor. The first data set can be regarded as a small-sized HSI, with the other two data sets being large-sized HSIs.

1) The Indian Pines image has 200 spectral reflectance bands and is of 145 × 145 pixels in size, with 21 025 pixels in total and a 20-m spatial resolution. A detailed description of this data can be referred to [9] and [10]. This image belongs to an agricultural scene, with 16 classes in total. It has a relatively concentrated and smooth land-cover distribution. Fig. 4 shows the false-color

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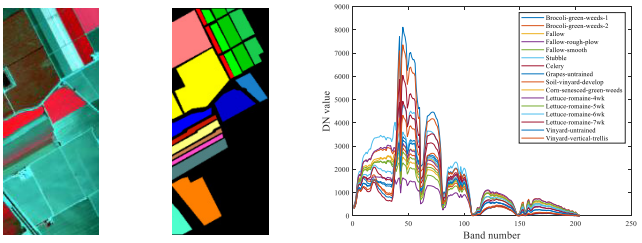



Fig. 5. Salinas image: (a) false-color image (R: 50, G: 27, and B: 17), (b) ground truth, and (c) mean spectra of the 16 classes.

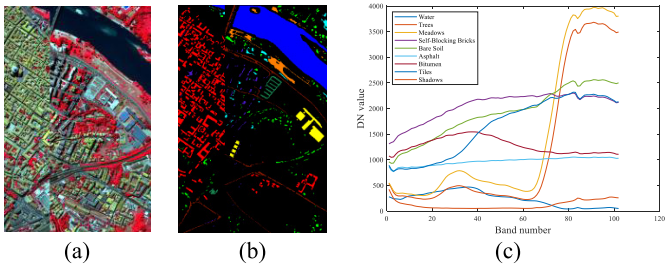



Fig. 6. Pavia Center image: (a) false-color image (R: 102, G: 56, and B: 31), (b) ground truth, and (c) mean spectra of the nine classes.

TABLE II
PARAMETER SETTINGS OF DIFFERENT CLUSTERING METHODS FOR THE SALINAS IMAGE

| Method | Parameters |
|---|---|
| FCM | $c=16$, $e=2$ |
| FCM-S1 | $c=16$, $e=2$, $\alpha=0.8$ |
| SSC-S | --- |
| L2-SSC | --- |
| LRSC | --- |
| SGCNR | $c=16$, $m=500$, $k=5$, $\rho$=1.1, $\mu=0.1$ |
| FSCAG | $c=16$, $m=1000$, $k=5$, $\alpha=0.8$ |
| SCC | $c=16$, $d=30$, $s=11$ |
| JSCC | $c=16$, $A=1637$, $d=25$, $s=9$ |

$c$ : the number of clusters; $e$ : fuzzy exponential; $\alpha$ : spatial regularization parameter; $\lambda$ : regularization parameter; $m$ : the number of anchors; $k$ : the number of neighbors; $\rho$ : penalty parameter; $\mu$ : penalty parameter; $d$ : the number of atoms for each category; $s$ : the sparsity level of the coefficient; $A$ : the number of super-pixels; $S$ : the size of the local window.

TABLE III
PARAMETER SETTINGS OF DIFFERENT CLUSTERING METHODS FOR THE PAVIA CENTER IMAGE

| Method | Parameters |
|---|---|
| FCM | $c=9$, $e=2$ |
| FCM-S1 | $c=9$, $e=2$, $\alpha=0.1$ |
| SSC-S | --- |
| L2-SSC | --- |
| LRSC | --- |
| SGCNR | $c=9$, $m=1000$, $k=5$, $\rho$=1.1, $\mu=0.1$ |
| FSCAG | $c=9$, $m=1000$, $k=5$, $\alpha=0.3$ |
| SCC | $c=9$, $d=30$, $s=9$ |
| JSCC | $c=9$, $A=9463$, $d=25$, $s=9$ |

$c$ : the number of clusters; $e$ : fuzzy exponential; $\alpha$ : spatial regularization parameter; $\lambda$ : regularization parameter; $m$ : the number of anchors; $k$ : the number of neighbors; $\rho$ : penalty parameter; $\mu$ : penalty parameter; $d$ : the number of atoms for each category; $s$ : the sparsity level of the coefficient; $A$ : the number of super-pixels; $S$ : the size of the local window.

image, the ground truth and the mean spectra of the 16 classes, from which it can be observed that most of the classes are very difficult to distinguish due to their high similarity. The optimal parameters of each method for this scene are given in Table I.

2) The Salinas image has 224 spectral bands and is of 512 × 217 pixels in size, with 111 104 pixels in total and a 3.7-m spatial resolution. A detailed description of this data can be referred to [9]. Similarly, this image also belongs to an agricultural scene, with a relatively concentrated and smooth land-cover distribution. In this scene, there are 16 classes in total, including many subclasses. The false-color image, the ground truth, and the mean spectra of the 16 classes are shown in Fig. 5(a)–(c), respectively. From Fig. 5(c), we can find that the spectra of many classes are seriously mixed, which leads to a large challenge to clustering. The optimal parameters of each method for this scene are given in Table II.
3) The Pavia Center image has 102 spectral bands and is of 1096 × 715 pixels in size, with 783 640 pixels in total and a 1.3-m spatial resolution. A detailed description of this data can be referred to [48]. Different from the former two validation images, this image belongs to a city center scene and contains nine different classes in total, with a relatively complex distribution, which leads to some difficulties in clustering. Fig. 6 shows the false-color image, the ground truth, and the mean spectra of the nine classes. The optimal parameters of each method for this scene are shown in Table III.

To comprehensively compare the performance of different clustering methods, both visual and quantitative results are provided. The quantitative indicators include producer's accuracy (PA), overall accuracy (OA), kappa, and purity, as well as the time cost. The average result of five independent Monte Carlo experiments is used as the final clustering result.

### B. Computational Complexity and Memory Cost Analysis

For the computational complexity, both FCM and FCM-S1 have a relatively low complexity of $O(MNDct)$, where $t$ stands

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TABLE IV
MEMORY COST OF THE SUBSPACE CLUSTERING METHODS FOR SAVING THE REPRESENTATION COEFFICIENT MATRIX

| Indicators | Indian Pines Image | Salinas Image | Pavia Center Image |
|---|---|---|---|
| The matrix size (Pixels) | $21025 \times 21025$ | $111104 \times 111104$ | $783640 \times 783640$ |
| The memory cost (GB) | 3.29 | 91.97 | 4575.34 |

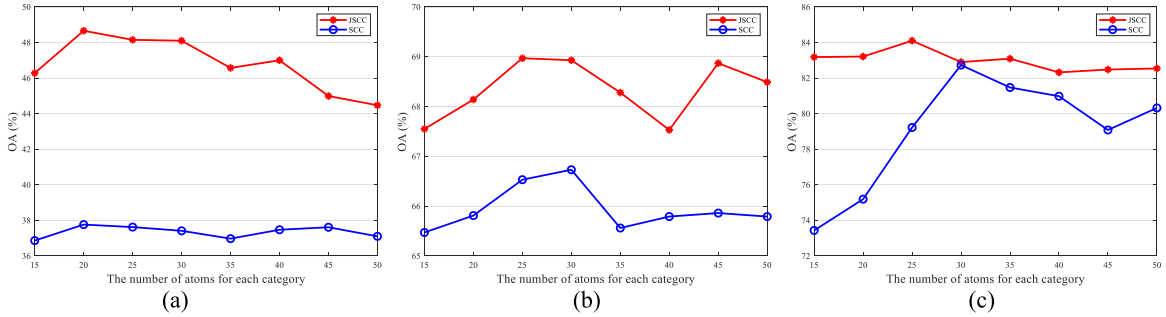



Fig. 7. Parameter analysis for $d$: OA versus the number of atoms for each category. (a) Indian Pines image. (b) Salinas image. (c) Pavia Center image.

for the number of iterations. By comparison, the subspace clustering methods, for example, SSC-S, L2-SSC, and LRSC, have a much larger complexity of $O[(MN)^3 t]$, which scales cubically with the number of pixels and is intolerable for practical applications. By introducing the anchor graph with much smaller size, the recently developed FSCAG and SGCNR methods have much lower complexity and are very efficient. Specifically, the complexity of FSCAG is $O(MNDm)$ [48], where $m$ denotes the number of anchors, and the complexity of SGCNR is $O(MND\log m + MNc^2 + MNck + c^3)$ [49], where $k$ is the number of nearest neighbors, with $m, k \ll MN$. Comparatively speaking, the computational complexity of the two sparsity-based models, i.e., SCC and JSCC, is $O(MNDKs)$, with $D, K, s \ll MN$. It can be noted that compared with the subspace clustering methods, the computational complexity of the two sparsity-based methods is reduced to a large degree, which makes them more attractive in practical applications and improves their applicability to large HSIs.

For the memory cost, except for the three subspace clustering methods, for example, SSC-S, L2-SSC, and LRSC, the other clustering methods discussed in this article are with low memory consumption and can be well-applied to large HSIs. Here, we focus on the memory cost of the subspace clustering methods. It should be noted that the representation coefficient matrix of the subspace clustering methods is at a very large size, i.e., the number of pixels multiplying the number of pixels, which is the basis of adjacency matrix construction and graph segmentation. Saving such a large matrix for subsequent computations will consume a very large memory, as illustrated in Table IV. From which, it can be seen that the subspace clustering methods are very memory-consuming, which greatly limits their applicability to large HSIs.

### C. Parameter Analysis

There are two common key parameters in the proposed SCC and JSCC algorithms: the dictionary size and the sparsity level of the coefficient. For JSCC, there is another important parameter, i.e., the number of super-pixels. The sensitivity and influence of each parameter were carefully analyzed, by fixing the other parameters at their optimal values. In addition, all three experimental images, i.e., the Indian Pines image, the Salinas image, and the Pavia Center image, are used to comprehensively analyze the impact of each parameter on the performance of SCC and JSCC.

*Dictionary Size:* The dictionary size is an important parameter of the proposed SCC and JSCC algorithms, which controls the sparse recovery accuracy and the quality of the structured dictionary. As we know, the redundancy of the dictionary and the computational complexity will significantly increase with the increment of the dictionary size. In practice, the dictionary size is determined by the number of atoms for each category, i.e., parameter $d$. Fig. 7 shows that the clustering precision changes with the value of parameter $d$.

From Fig. 7, we can find that parameter $d$ has a certain influence on the performance of the proposed algorithms. Both JSCC and SCC cannot perform well when $d$ is very small. This is because when the number of atoms for each category is very small, the collaboration and recovery capability of the structured dictionary is decreased to a large degree, and the sparse recovery precision cannot be guaranteed. With the increment of $d$, the clustering accuracy of both JSCC and SCC increases first and then stabilizes, to a certain degree. In general, the performance of the proposed algorithms is relatively stable over a large range of parameter $d$ for all three applications, and it can be easily fine-tuned in practice.

*Sparsity Level of the Coefficient:* The sparsity level of the coefficient, i.e., parameter $s$, is also very important for the proposed algorithms, as it directly affects the precision of the sparse recovery and the discrimination of the learned coefficients. Fig. 8 shows that the clustering precision changes with the value of this parameter.

From this, we can find that the sparsity level of the coefficient can directly affect the final clustering accuracy of JSCC and SCC. Generally speaking, the performance of both

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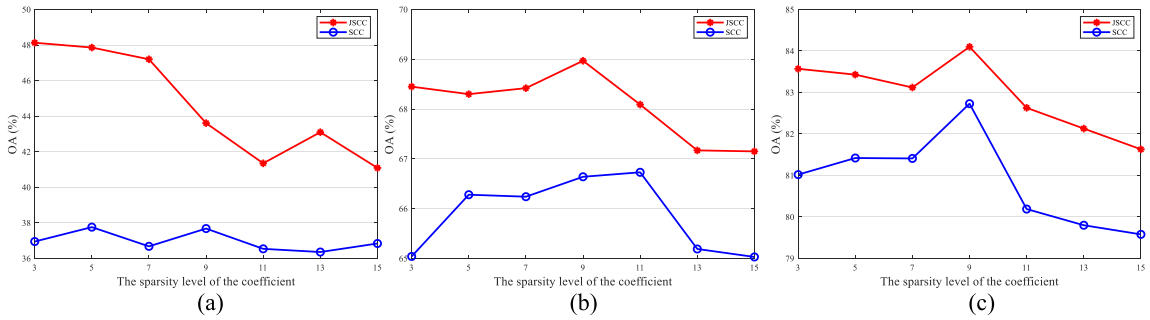



Fig. 8. Parameter analysis for $s$: OA versus the sparsity level of the coefficient. (a) Indian Pines image. (b) Salinas image. (c) Pavia Center image.

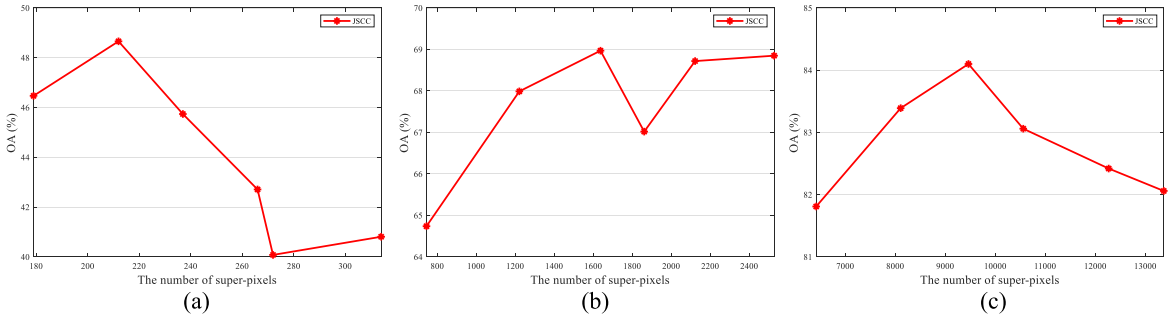



Fig. 9. Parameter analysis for $A$: OA versus the number of super-pixels. (a) Indian Pines image. (b) Salinas image. (c) Pavia Center image.

JSCC and SCC is relatively stable for all three experimental images, and parameter $s$ can be easily fine-tuned in practical applications. By fully exploring the interpixel correlation of HSIs, JSCC can obtain a better performance than SCC, in most cases.

*Number of Super-Pixels:* The number of super-pixels, i.e., parameter $A$, is another important parameter of the proposed JSCC algorithm, which determines the neighborhood size and the intensity of the spatial information and has a direct influence on the final clustering accuracy. Fig. 9 shows that the clustering precision changes with the value of parameter $A$. From this, we can easily find that this parameter occupies an important position in the clustering performance of JSCC. Although for different HSI data sets, the optimal value of parameter $A$ is different, which is generally related to the complexity of the spatial texture and the size of the image, the performance of JSCC is relatively stable over a large range of this parameter. Therefore, in practice, an optimal $A$ can be easily found for a specific HSI data.

### *D. Experimental Results for the AVIRIS Indian Pines Image*

The first experiment was with the well-known AVIRIS Indian Pines image, with the cluster maps and the quantitative evaluation provided in Fig. 10 and Table V, respectively. From the experimental results, a conclusion can be drawn that, overall, the spectral–spatial methods obtain better clustering effect compared with the spectral-based methods. The proposed SCC algorithm outperforms the other spectral-based algorithms, and the JSCC algorithm obtains the best clustering performance.

Limited by the $n$-sphere structure assumption, which cannot be well-satisfied by HSIs, the centroid-based methods fail to obtain satisfactory clustering results. Specifically, FCM performs poorly in this scene to obtain the lowest OA of 31.35%, with many classes misclassified and a lot of noise presented in the cluster map. Especially for the Soybean-no till class, a large proportion of it is misclassified as Soybean-minimal-till or Corn-minimum-till. As a result, its PA is only 23.15%. By fully considering the spatial information of HSIs, FCM-S1 obtains a better clustering performance, with a smoother cluster map. For Grass/Pasture and Hay-windrowed, the PAs are improved from 44.31% and 86.61% to 46.29% and 98.83%, respectively. However, the clustering result of FCM-S1 still contains a high level of misclassification, which leads to a relatively low OA, i.e., 32.70%, and is far from satisfactory.

Compared with the classical methods, the subspace clustering algorithms perform much better in this scene. Specifically, LRSC performs well for the Alfalfa and the Soybean-minimum-till classes to obtain a higher OA of 36.68%. However, because of ignoring spatial information, there is still a lot of misclassification and noise in the cluster map, with several classes failed to be recognized. By comparison, SSC-S and L2-SSC obtain much better clustering results by simultaneously exploring the spectral–spatial information of HSIs, with smoother maps and higher OAs of 43.37% and 43.11%, respectively. For several classes, i.e., Grass/Pasture, Grass/Trees, and Soybean-clean, these two methods achieve the best and second-best precisions among all the methods. In general, SSC-S and L2-SSC achieve the second-best and the third-best accuracy, which suggests their effectiveness and superiority. However, it should be noted that the time costs of these subspace clustering methods are very high, which greatly reduces their attractiveness in practical applications. In contrast, the two recently developed anchor-graph-based methods for large HSIs, i.e., SGCNR and FSCAG, signif-

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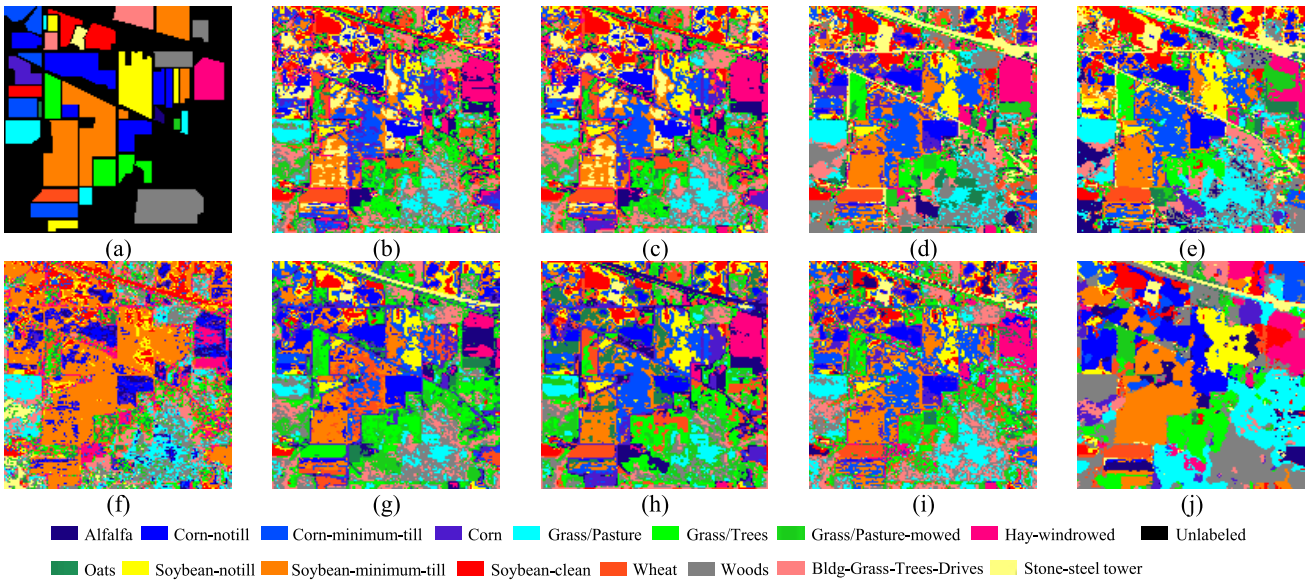



Fig. 10. Cluster maps obtained by different methods for the Indian Pines image. (a) Ground truth. (b) FCM. (c) FCM-S1. (d) SSC-S. (e) L2-SSC. (f) LRSC. (g) SGCNR. (h) FSCAG. (i) SCC. (j) JSCC.

TABLE V
QUANTITATIVE EVALUATION OF THE CLUSTERING RESULTS FOR THE INDIAN PINES IMAGE

| Method | Class | FCM | FCM-S1 | SSC-S | L2-SSC | LRSC | SGCNR | FSCAG | SCC | JSCC |
|---|---|---|---|---|---|---|---|---|---|---|
| PA (%) | 1 | 39.13 | 15.22 | 0 | 0 | **52.17** | 26.52 | 9.57 | 8.26 | 13.48 |
| | 2 | 25.63 | 26.58 | 26.75 | **42.58** | 27.31 | 36.13 | 23.85 | 27.87 | 33.40 |
| | 3 | 24.58 | 26.10 | 36.87 | 18.80 | 1.81 | 22.80 | 34.36 | **41.47** | 30.17 |
| | 4 | 6.33 | 13.92 | 10.13 | 0 | 13.50 | 13.00 | **24.47** | 9.62 | 16.12 |
| | 5 | 44.31 | 46.29 | **62.73** | 62.32 | 57.97 | 44.68 | 40.08 | 46.00 | 57.93 |
| | 6 | 26.99 | 29.78 | 72.05 | **80.41** | 32.47 | 44.08 | 39.45 | 52.41 | 55.97 |
| | 7 | 0 | 0 | 0 | **89.29** | 0 | 14.29 | 0 | 12.86 | 0.71 |
| | 8 | 86.61 | 98.83 | **100** | 54.39 | 28.03 | 66.78 | 92.59 | 78.62 | 72.51 |
| | 9 | 20.00 | **29.00** | 0 | 25.00 | 25.00 | 12.00 | 9.00 | 5.00 | 15.00 |
| | 10 | 23.15 | 25.60 | 35.08 | 46.81 | 17.18 | 34.75 | 29.12 | 27.65 | **48.66** |
| | 11 | 28.19 | 26.66 | 37.68 | 37.19 | **69.86** | 34.60 | 30.22 | 39.41 | 58.71 |
| | 12 | 23.61 | 24.72 | 30.69 | **31.53** | 19.90 | 15.82 | 22.56 | 12.65 | 19.26 |
| | 13 | **99.02** | 98.73 | **99.02** | 95.61 | 23.90 | 74.54 | 87.71 | 87.12 | 61.27 |
| | 14 | 34.16 | 32.98 | 49.33 | 45.69 | 41.03 | 44.36 | 37.45 | 38.62 | **70.31** |
| | 15 | 17.62 | 18.81 | 15.54 | 15.28 | 19.17 | 15.60 | 17.67 | 19.07 | **25.49** |
| | 16 | 59.14 | 58.06 | **97.85** | 94.62 | 0 | 59.14 | 65.81 | 69.03 | 37.63 |
| OA (%) | | 31.35 | 32.70 | 43.37 | 43.11 | 36.68 | 36.31 | 34.70 | 37.76 | **48.66** |
| Kappa | | 0.2561 | 0.2695 | 0.3757 | 0.3667 | 0.2713 | 0.2946 | 0.2887 | 0.3091 | **0.4254** |
| Purity | | 0.5015 | 0.5082 | 0.5588 | 0.5670 | 0.4571 | 0.5105 | 0.5137 | 0.5222 | **0.5689** |
| Time (s) | | 299.98 | 381 | $2.90\times10^4$ | $2.80\times10^4$ | $7.67\times10^3$ | 9.05 | **1.33** | 13.94 | 11.16 |

1: Alfalfa; 2: Corn-notill; 3: Corn-minimum-till; 4: Corn; 5: Grass/Pasture; 6: Grass/Trees; 7: Grass/Pasture-mowed; 8: Hay-windrowed; 9: Oats; 10: Soybean-notill; 11: Soybean-minimum-till; 12: Soybean-clean; 13: Wheat; 14: Woods; 15: Bldg-Grass-Trees-Drives; 16: Stone-steel towers.

icantly reduce the time cost, with the running times being only 9.05 and 1.33 s, respectively. However, it should be noted that the clustering accuracy of these two methods is not satisfactory, and they obtain OAs of only 36.31% and 34.70%, respectively.

Let us turn to the two proposed sparsity-based clustering algorithms, i.e., SCC and JSCC. SCC obtains an OA of 37.76% in this scene, which is the best among the results of spectral-based methods. To be specific, compared with SGCNR, SCC performs better, with an increment in OA of about 1.5%. For example, the PAs for the Corn-minimum-till and the Soybean-minimum-till classes are improved from 22.80% and 34.60% to 41.47% and 39.41%, respectively. However, because of the absence of spatial information, SCC still cannot get rid of the interferences from misclassification and noise. Compared with SCC, JSCC improves the clustering performance to a large degree, by fully exploring the interpixel correlation and promoting the neighboring pixels to be jointly

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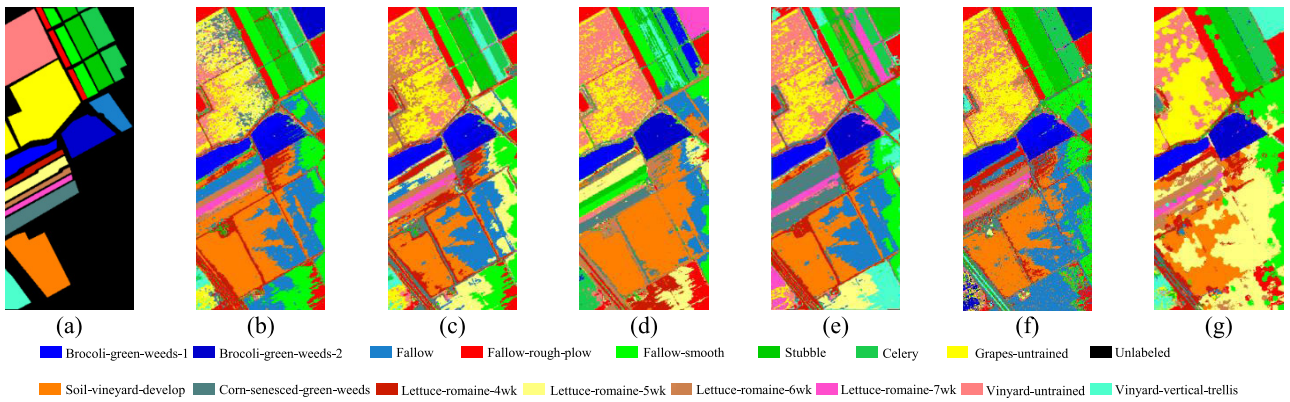



Fig. 11. Cluster maps obtained by different methods for the Salinas image. (a) Ground truth. (b) FCM. (c) FCM-S1. (d) SGCNR. (e) FSCAG. (f) SCC. (g) JSCC.

sparse coded. It effectively reduces the misclassification and promotes the smoothness of the cluster map. Consequently, JSCC obtains the best quantitative evaluation results, including the highest OA and kappa of 48.66% and 0.4254, respectively. Overall, JSCC exhibits certain superiority to the other comparative methods in this scene, which demonstrates its effectiveness.

The purity index (which ranges from 0 to 1) is used to further evaluate the clustering effect of different methods. The larger the purity, the better the clustering effect [69]. From the penultimate row of Table II, we can find that the proposed SCC and JSSC algorithms again show some superiority in this indicator, to obtain higher purity values of 0.5222 and 0.5689, respectively. On the other hand, it is also worth mentioning that compared with the subspace clustering methods, i.e., SSC-S, L2-SSC, and LRSC, the time costs of SCC and JSCC are significantly reduced, from $2.9 \times 10^4$, $2.8 \times 10^4$, and $7.67 \times 10^3$ s to only 13.94 and 11.64 s, respectively, which is comparable to the other efficient methods, i.e., SGCNR and FSCAG, and faster than FCM and FCM-S1. For JSCC, the running time is the sum of the time costs of the segmentation process and the clustering process.

### *E. Experimental Results for the AVIRIS Salinas Image*

The second experiment was with a large HSI, i.e., the Salinas image, with the cluster maps and the quantitative evaluation provided in Fig. 11 and Table VI, respectively. Similar conclusions can be drawn from the experimental results. Overall, the spectral–spatial methods again outperform the spectral-based methods.

Specifically, FCM performs worst again in this scene, and its clustering result is seriously plagued by misclassification and noise. The OA of FCM is at a very low level, i.e., 53.51%. With the help of the spatial information, FCM-S1 obtains a much better performance to get a smoother cluster map, with an increment in OA of about 9%. For Lettuce-romaine-4wk, FCM-S1 effectively improves the PA from 24.85% to 75.34%.

Unfortunately, due to the high computational complexity and large memory cost, the subspace clustering methods, i.e., SSC-S, L2-SSC, and LRSC, suffer from the "out of memory" error and cannot work in this scene. In contrast, SGCNR and FSCAG can effectively cluster this large HSI, at a time cost of only 57.93 and 13.02 s, respectively, which demonstrates the scalability and efficiency of such methods. However, the clustering precisions of these two methods are still at a relatively low level, with OAs of only 64.84% and 64.95%, respectively.

In comparison, the proposed SCC algorithm achieves a good performance. It well-distinguishes different classes to obtain the second-best clustering accuracy, with an OA of 66.73%. Compared with SCC, JSCC further enhances the clustering performance to obtain a smoother cluster map with the highest accuracy, i.e., an OA of 68.97%, a kappa of 0.6548, and a purity of 0.7115, which further confirms its effectiveness. However, it fails to recognize some classes well, such as the Fallow class, which is a defect to be overcome. On the other hand, the time costs of SCC and JSCC are 163.34 and 270.72 s, respectively, which are less than those of FCM and FCM-S1. This demonstrates the applicability of the two sparsity-based clustering algorithms in large HSI clustering to some degree.

### *F. Experimental Results for the ROSIS Pavia Center Image*

The third experiment was with a very large-sized HSI, i.e., ROSIS Pavia Center image, with the cluster maps and the quantitative evaluation provided in Fig. 12 and Table VII, respectively. In this experiment, the proposed JSCC and SCC algorithms again achieve the best and the second-best clustering performance, which further confirms their effectiveness. In addition, the spectral–spatial methods again show a large superiority to the spectral-based methods. Specifically, FCM-S1 obviously outperforms FCM, obtaining a much smoother cluster map with a higher precision. FSCAG is also superior to SGCNR, to obtain a much higher clustering precision. Similarly, JSCC obtains better performance com-

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TABLE VI
QUANTITATIVE EVALUATION OF THE CLUSTERING RESULTS FOR THE SALINAS IMAGE

| Method | Class | FCM | FCM-S1 | SGCNR | FSCAG | SCC | JSCC |
|---|---|---|---|---|---|---|---|
| PA (%) | 1 | **99.75** | 99.68 | 73.93 | 57.93 | 79.86 | 39.96 |
| | 2 | 39.32 | 50.05 | 80.37 | **96.42** | 56.81 | 94.43 |
| | 3 | **56.22** | 35.32 | 44.86 | 53.16 | 34.46 | 7.64 |
| | 4 | 99.24 | 98.22 | 69.73 | 58.06 | 95.54 | **99.80** |
| | 5 | **91.04** | 84.97 | 83.86 | 90.46 | 84.15 | 68.43 |
| | 6 | 86.29 | 95.36 | 86.69 | 83.86 | 96.18 | **96.95** |
| | 7 | 90.49 | 83.30 | 80.60 | 44.72 | 98.43 | **99.22** |
| | 8 | 36.43 | 50.46 | 55.70 | 57.57 | 72.60 | **77.24** |
| | 9 | 72.29 | 86.84 | 79.06 | **92.91** | 82.10 | 77.89 |
| | 10 | 4.92 | 1.83 | 63.97 | **84.83** | 40.36 | 11.60 |
| | 11 | 24.85 | 75.34 | 27.60 | 62.38 | 54.25 | **75.62** |
| | 12 | 28.07 | **62.41** | 58.26 | 0.25 | 9.48 | 51.03 |
| | 13 | 98.89 | **99.17** | 59.04 | 0 | 96.07 | 93.14 |
| | 14 | 88.02 | 86.34 | 55.08 | **91.05** | 88.13 | 82.34 |
| | 15 | 30.95 | 52.86 | 56.13 | **62.10** | 42.54 | 51.91 |
| | 16 | 14.14 | 5.94 | 16.17 | 0.46 | 29.13 | **57.17** |
| OA (%) | | 53.51 | 62.61 | 64.84 | 64.95 | 66.73 | **68.97** |
| Kappa | | 0.4948 | 0.5894 | 0.6101 | 0.6121 | 0.6303 | **0.6548** |
| Purity | | 0.6660 | 0.6710 | 0.6903 | 0.6860 | 0.6823 | **0.7115** |
| Time (s) | | $1.02\times10^3$ | $3.01\times10^3$ | 57.93 | **13.02** | 163.34 | 270.72 |

1: Brocoli-green-weeds-1; 2: Brocoli-green-weeds-2; 3: Fallow; 4: Fallow-rough-plow; 5: Fallow-smooth; 6: Stubble; 7: Celery; 8: Grapes-untrained; 9: Soil-vinyard-develop; 10: Corn-senesced-green-weeds; 11: Lettuce-romaine-4wk; 12: Lettuce-romaine-5wk; 13: Lettuce-romaine-6wk; 14: lettuce-romaine-7wk; 15: Vinyard-untrained; 16: Vinyard-vertical-trellis.

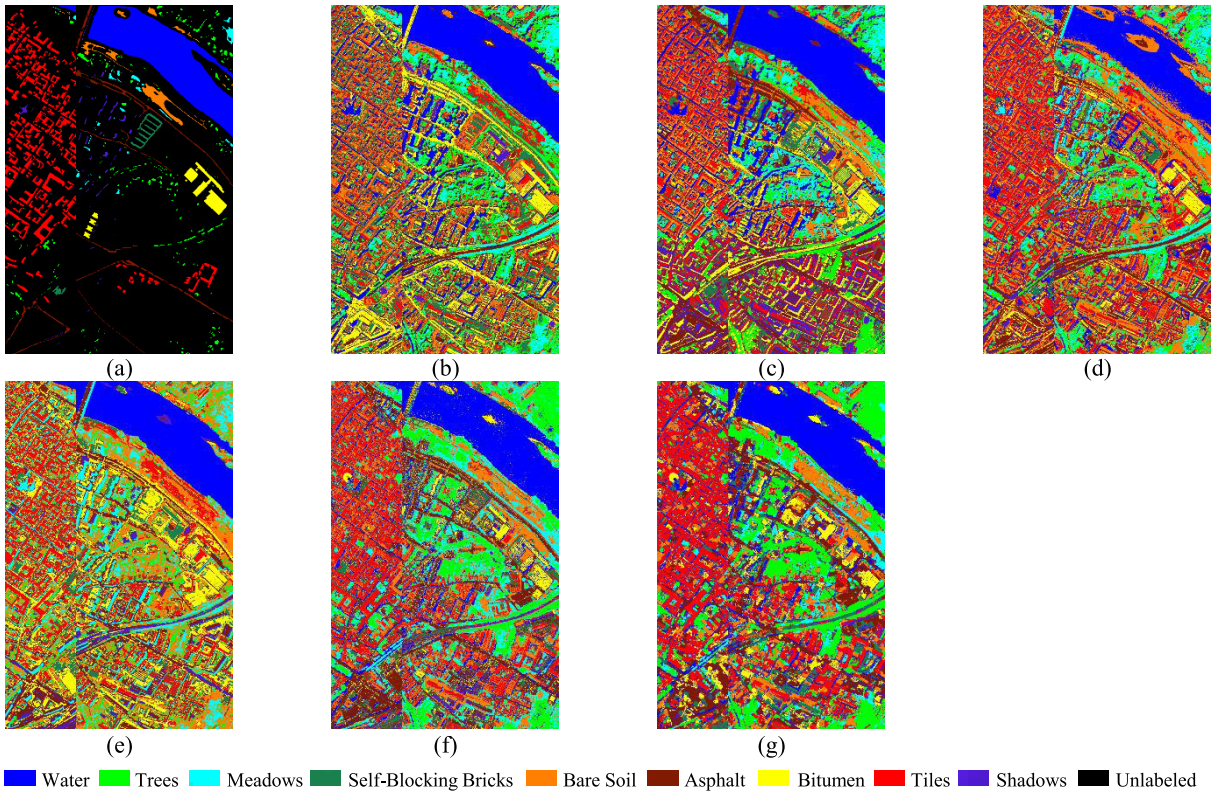



Fig. 12. Cluster maps obtained by different methods for the Pavia Center image. (a) Ground truth. (b) FCM. (c) FCM-S1. (d) SGCNR. (e) FSCAG. (f) SCC. (g) JSCC.

pared with SCC, with a smoother cluster map and a higher OA and kappa of 84.10% and 0.7754, respectively. It can be noted that although JSCC yields a smooth clustering result and some details may be lost, it performs well for most classes, with the boundary and spatial details of ground objects relatively well-preserved. In general, JSCC may perform better on HSIs with

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TABLE VII
QUANTITATIVE EVALUATION OF THE CLUSTERING RESULTS FOR THE PAVIA CENTER IMAGE

| Method | Class | FCM | FCM-S1 | SGCNR | FSCAG | SCC | JSCC |
|---|---|---|---|---|---|---|---|
| PA (%) | 1 | 99.09 | **99.27** | 93.42 | 98.56 | 95.53 | 97.73 |
| | 2 | 35.69 | 65.43 | 36.44 | 41.36 | 89.99 | **94.92** |
| | 3 | **82.72** | 69.51 | 44.50 | 62.17 | 42.33 | 29.54 |
| | 4 | 10.43 | **77.41** | 36.40 | 39.20 | 50.78 | 60.28 |
| | 5 | 49.39 | 28.75 | 20.76 | 27.63 | 44.73 | **55.10** |
| | 6 | 13.01 | 81.51 | 74.82 | **89.46** | 70.64 | 41.69 |
| | 7 | **78.41** | 73.41 | 54.58 | 57.74 | 72.63 | 50.68 |
| | 8 | 34.00 | 49.79 | 51.30 | 58.03 | 81.63 | **90.95** |
| | 9 | 0 | 1.97 | **67.13** | 40.11 | 10.63 | 8.87 |
| OA (%) | | 64.56 | 74.82 | 69.46 | 75.22 | 82.73 | **84.10** |
| Kappa | | 0.5236 | 0.6559 | 0.5905 | 0.6629 | 0.7594 | **0.7754** |
| Purity | | 0.8308 | 0.8559 | 0.8220 | 0.8552 | **0.8613** | 0.8578 |
| Time (s) | | $3.85\times10^3$ | $4.68\times10^3$ | 495.40 | **128.87** | $2.98\times10^3$ | $1.33\times10^3$ |

1: Water; 2: Trees; 3: Meadows; 4: Self-Blocking Bricks; 5: Bare Soil; 6: Asphalt; 7: Bitumen; 8: Tiles; 9: Shadows.

a homogeneous land-cover distribution owing to the spatial help of super-pixels, but it also works on HSIs with many heterogeneous regions.

Furthermore, the time costs of SCC and JSCC for this large scene are $2.98 \times 10^3$ and $1.33 \times 10^3$s, respectively, which are lower than those of FCM and FCM-S1, and are acceptable for practical applications.

## V. CONCLUSION

In this article, for the first time, we have attempted to explore the potential of sparse coding and the recovery residual in HSI clustering, and two novel sparsity-based clustering algorithms have been proposed for large HSIs. Specifically, based on a structured dictionary constructed by $k$-means and KNN in the case of no labeled samples available, an SCC model is constructed to cluster HSIs according to the recovery residual minimization criterion. Unlike the subspace clustering methods, SCC invokes a pixel-wise sparse recovery problem instead of the large-scale graph optimization problem of the whole image to reduce the computational complexity and the time-and-memory cost, which improves its practicability to a large degree. Given the drawback that SCC ignores the spatial information, through using super-pixel segmentation to produce homogeneous neighborhoods, a JSCC model is constructed to better explore the interpixel correlation of HSIs and further improve the clustering performance. The experimental results illustrate that the SCC and JSCC algorithms outperform the other state-of-the-art clustering methods and have a large potential for large HSI clustering.

However, there is still some improvement room. First, through more efficiently and more accurately exploring the correlation between neighboring pixels and integrating the spatial information in a more precise way, the clustering performance of the proposed algorithms could be further enhanced. In addition, the performance and the practicability of the proposed algorithms could be enhanced by automatically searching the optimal parameters. These will be our future research lines.

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